

CODSOFT CYBER SECURITY INTERNSHIP

TASK 3 — SECURE CODE ASSESSMENT

Python User Registration and Login Application using SQLite

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Internship: Cyber Security Internship

Task: 3

Technology: Python + SQLite

1. INTRODUCTION

The objective of this task is to examine an application source code, identify common security weaknesses and coding flaws, and implement secure coding practices to improve the application's security.

For this assessment, a Python user registration and login application using SQLite was selected. The source code was manually reviewed to identify security weaknesses. After identifying the weaknesses, a secure version of the application was prepared by applying appropriate security practices.

2. APPLICATION DESCRIPTION

The application selected for this assessment is a Python User Registration and Login Application using SQLite.

The application provides basic user registration and login functionality. SQLite is used as the database for storing user information.

Two versions were prepared: Vulnerable Version and Secure Version. The vulnerable version was reviewed to identify security weaknesses, while the secure version was improved by implementing secure coding practices.

3. VULNERABLE CODE ANALYSIS

The vulnerable version of the application was manually reviewed to identify security weaknesses and coding flaws.

The main security weakness identified in the application was related to password storage.

3.1 INSECURE PASSWORD STORAGE

In the vulnerable implementation, passwords were handled using an insecure password-storage approach.

Storing passwords insecurely can expose user credentials if the database is accessed by an unauthorized person.

Risk: High

Security Impact: If an attacker gains access to the database, insecurely stored passwords could be

exposed and potentially used to access user accounts.

4. RECOMMENDED SECURITY FIX

The identified password-storage weakness was addressed in the secure version of the application.

Instead of storing passwords insecurely, the secure implementation uses a secure password-hashing approach.

Recommended practices:

- Do not store passwords as plain text.
- Use a secure password-hashing mechanism.
- Verify passwords securely during login.
- Protect the user database from unauthorized access.

5. SECURE CODE IMPLEMENTATION

After reviewing the vulnerable version, a secure version of the application was prepared. The secure version improves password protection by using a secure hashing approach for stored passwords.

The application continues to provide registration and login functionality while improving the security of user credentials.

6. VULNERABILITY AND FIX SUMMARY

Vulnerability: Insecure Password Storage

Description: Passwords were handled using an insecure storage method in the vulnerable version.

Recommended Fix: Use secure password hashing and verify passwords securely during login.

7. TESTING AND OUTPUT

The secure application was executed to verify that the implemented changes work correctly.

Testing performed:

- User registration was tested.
- User login functionality was tested.
- The secure version was executed successfully.
- Application output was checked after the security improvement.

Screenshots of the vulnerable code, secure code, and application output can be included as evidence of the work.

8. PROJECT FILES

The Task 3 project contains the vulnerable and secure Python source code, requirements.txt, the SQLite database file, the Secure Code Assessment Report, and the project demonstration video.

9. CONCLUSION

The Python user registration and login application was examined as part of the Secure Code Assessment. The vulnerable version was manually reviewed and a security weakness related to

password storage was identified. A secure version was then prepared using a secure password-hashing approach.

This task demonstrates the importance of reviewing source code for security weaknesses and applying secure coding practices to protect user information.

10. DECLARATION

I hereby confirm that this Secure Code Assessment was prepared as part of the CodSoft Cyber Security Internship — Task 3. The application was reviewed, security weaknesses were identified, and security improvements were implemented as part of this assessment.

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